

Unbiased Estimation

Example 3.1 shows that a clean comparison between two estimators is not always possible: if their risk functions cross, one estimator will be preferable for θ in some subset of the parameter space Ω , and the other will be preferable in a different subset of Ω . In some cases this problem will not arise if both estimators are unbiased. We may then be able to identify a best unbiased estimator. These ideas and limitations of the theory are discussed in Sections 4.1 and 4.2. Sections 4.3 and 4.4 concern distribution theory and unbiased estimation for the normal one-sample problem in which data are i.i.d. from a normal distribution. Sections 4.5 and 4.6 introduce Fisher information and derive lower bounds for the variance of unbiased estimators.

4.1 Minimum Variance Unbiased Estimators

An estimator δ is called *unbiased* for $g(\theta)$ if

$$E_{\theta}\delta(X) = g(\theta), \quad \forall \theta \in \Omega. \quad (4.1)$$

If an unbiased estimator exists, g is called *U-estimable*.

Example 4.1. Suppose X has a uniform distribution on $(0, \theta)$. Then δ is unbiased if

$$\int_0^{\theta} \delta(x)\theta^{-1} dx = g(\theta), \quad \forall \theta > 0,$$

or if

$$\int_0^{\theta} \delta(x) dx = \theta g(\theta), \quad \forall \theta > 0. \quad (4.2)$$

So g cannot be U-estimable unless $\theta g(\theta) \rightarrow 0$ as $\theta \downarrow 0$. If g' exists, then differentiating (4.2), by the fundamental theorem of calculus,

$$\delta(x) = \frac{d}{dx}(xg(x)) = g(x) + xg'(x).$$

For instance, if $g(\theta) = \theta$, $\delta(X) = 2X$.

Example 4.2. If X has the binomial distribution with n trials and success probability θ , and if $g(\theta) = \sin \theta$, then δ will be unbiased if

$$\sum_{k=0}^n \delta(k) \binom{n}{k} \theta^k (1-\theta)^{n-k} = \sin \theta, \quad \forall \theta \in (0, 1).$$

The left hand side of this equation is a polynomial in θ with degree at most n . The sine function cannot be written as a polynomial, therefore $\sin \theta$ is not U-estimable.

With squared error loss, $L(\theta, d) = (d - g(\theta))^2$, the risk of an unbiased estimator δ is

$$R(\theta, \delta) = E_{\theta}(\delta(X) - g(\theta))^2 = \text{Var}_{\theta}(\delta(X)),$$

and so the goal is to minimize the variance.

Definition 4.3. An unbiased estimator δ is uniformly minimum variance unbiased (UMVU) if

$$\text{Var}_{\theta}(\delta) \leq \text{Var}_{\theta}(\delta^*), \quad \forall \theta \in \Omega,$$

for any competing unbiased estimator δ^* .

In a general setting there is no reason to suspect that there will be a UMVU estimator. However, if the family has a complete sufficient statistic, a UMVU will exist, at least when g is U-estimable.

Theorem 4.4. Suppose g is U-estimable and T is complete sufficient. Then there is an essentially unique unbiased estimator based on T that is UMVU.

Proof. Let $\delta = \delta(X)$ be any unbiased estimator and define

$$\eta(T) = E[\delta|T],$$

as in the Rao–Blackwell theorem (Theorem 3.28). By smoothing,

$$g(\theta) = E_{\theta}\delta = E_{\theta}E_{\theta}[\delta|T] = E_{\theta}\eta(T),$$

and thus $\eta(T)$ is unbiased. Suppose $\eta^*(T)$ is also unbiased. Then

$$E_{\theta}[\eta(T) - \eta^*(T)] = 0, \quad \forall \theta \in \Omega,$$

and by completeness, $\eta(T) - \eta^*(T) = 0$ (a.e. $\mathcal{P}$). This shows that the estimator $\eta(T)$ is essentially unique; any other unbiased estimator based on T will equal $\eta(T)$ except on a $\mathcal{P}$ -null set. The estimator $\eta(T)$ has minimum variance by the Rao–Blackwell theorem with squared error loss. Specifically, if δ^* is any unbiased estimator, then $\eta^*(T) = E_{\theta}(\delta^*|T)$ is unbiased by the calculation above. With squared error loss, the risk of δ^* or $\eta^*(T)$ is the variance, and so

$$\text{Var}_{\theta}(\delta^*) \geq \text{Var}_{\theta}(\eta^*(T)) = \text{Var}_{\theta}(\eta(T)), \quad \forall \theta \in \Omega.$$

Thus, $\eta(T)$ is UMVU. □

From the uniqueness assertion in this theorem, if T is complete sufficient and $\eta(T)$ is unbiased, $\eta(T)$ must be UMVU. Viewing (4.1) as an equation for δ , any solution of the form $\delta = \eta(T)$ will be UMVU. This approach provides one strategy to find these estimators.

Example 4.5. Let $X_1, \dots, X_n$ be i.i.d. from the uniform distribution on $(0, \theta)$. From Example 3.16, $T = \max\{X_1, \dots, X_n\}$ is complete and sufficient for the family of joint distributions. Suppose $\eta(T)$ is unbiased for $g(\theta)$. Then

$$\int_0^\theta \eta(t) \frac{nt^{n-1}}{\theta^n} dt = g(\theta), \quad \theta > 0,$$

which implies

$$n \int_0^\theta t^{n-1} \eta(t) dt = \theta^n g(\theta), \quad \theta > 0.$$

If g is differentiable and $\theta^n g(\theta) \rightarrow 0$ as $\theta \downarrow 0$, then differentiation with respect to θ gives

$$n\theta^{n-1}\eta(\theta) = \frac{d}{d\theta}(\theta^n g(\theta)),$$

and so

$$\eta(t) = \frac{1}{nt^{n-1}} \frac{d}{dt}(t^n g(t)) = g(t) + \frac{tg'(t)}{n}, \quad t > 0.$$

When g is a constant c this argument shows the $\eta(T)$ must also equal c , and so T is complete.

When g is the identity function, $g(\theta) = \theta$, $\eta(t) = (n+1)t/n$. Thus, $(n+1)T/n$ is the UMVU of θ . Another unbiased estimator is $\delta = 2\bar{X}$. By the theory we have developed, $\eta(T)$ must have smaller variance than δ . In this example the comparison can be done explicitly. Since

$$E_\theta T^2 = \int_0^\theta t^2 \frac{nt^{n-1}}{\theta^n} dt = \frac{n}{n+2} \theta^2$$

and

$$E_\theta \eta^2(T) = \left(\frac{n+1}{n}\right)^2 E_\theta T^2 = \frac{(n+1)^2}{n(n+2)} \theta^2,$$

we have

$$\text{Var}_\theta(\eta(T)) = E_\theta \eta^2(T) - (E_\theta \eta(t))^2 = \frac{(n+1)^2}{n(n+2)} \theta^2 - \theta^2 = \frac{\theta^2}{n(n+2)}.$$

When $n = 1$, $T = 2X_1$, and so this formula implies that $\text{Var}_\theta(2X_i) = \theta^2/3$. Because δ is an average of these variables,

$$\text{Var}_\theta(\delta) = \frac{\theta^2}{3n}.$$

The ratio of the variance of $\eta(T)$ to the variance of δ is

$$\frac{\text{Var}_\theta(\eta(T))}{\text{Var}_\theta(\delta)} = \frac{3}{n+2}.$$

As $n \rightarrow \infty$ this ratio tends to zero, and so $\eta(T)$ is much more accurate than δ when n is large.

The proof of Theorem 4.4 also suggests another way to find UMVU estimators. If δ is an arbitrary unbiased estimator, then $\eta(T) = E[\delta|T]$ will be UMVU. So if any unbiased estimator can be identified, the UMVU can be obtained by computing its conditional expectation.

Example 4.6. Let $X_1, \dots, X_n$ be i.i.d. Bernoulli variables with $P_\theta(X_i = 1) = \theta = 1 - P_\theta(X_i = 0)$, $i = 1, \dots, n$. The marginal mass function can be written as $\theta^x(1 - \theta)^{1-x}$, $x = 0$ or 1 , and so the joint mass function is

$$\prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{T(x)} (1 - \theta)^{n-T(x)},$$

where $T(x) = x_1 + \dots + x_n$. These joint mass functions form an exponential family with

$$T = T(X) = X_1 + \dots + X_n \sim \text{Binomial}(\theta, n)$$

as a complete sufficient statistic. Consider unbiased estimation of $g(\theta) = \theta^2$. One unbiased estimator is $\delta = X_1 X_2$. The UMVU estimator must be $\eta(T) = E_\theta[X_1 X_2 | T] = P_\theta(X_1 = X_2 = 1 | T)$. Because

$$\begin{aligned} P_\theta(X_1 = X_2 = 1, T = t) &= P_\theta\left(X_1 = X_2 = 1, \sum_{i=3}^n X_i = t - 2\right) \\ &= \theta^2 \binom{n-2}{t-2} \theta^{t-2} (1 - \theta)^{n-t}, \end{aligned}$$

$$\begin{aligned} P_\theta(X_1 = X_2 = 1 | T = t) &= \frac{P_\theta(X_1 = X_2 = 1, T = t)}{P_\theta(T = t)} \\ &= \frac{\theta^2 \binom{n-2}{t-2} \theta^{t-2} (1 - \theta)^{n-t}}{\binom{n}{t} \theta^t (1 - \theta)^{n-t}} = \frac{t}{n} \frac{t-1}{n-1}. \end{aligned}$$

So $T(T-1)/(n^2 - n)$ is the UMVU estimator of θ^2 .

4.2 Second Thoughts About Bias

Although the approach developed in the previous section often provides reasonable estimators, the premise that one should only consider unbiased estimators is suspect. Estimators with considerable bias may not be worth considering, but estimators with small bias may be quite reasonable.

Example 4.1, continued. As before, $X_1, \dots, X_n$ are i.i.d. from the uniform distribution on $(0, \theta)$, and $T = \max\{X_1, \dots, X_n\}$ is complete sufficient. The UMVU estimator $(n+1)T/n$ is a multiple of T , but is it the best multiple of T ? To address this question, let us calculate the risk of $\delta_a = aT$ under squared error loss. From our prior calculations,

$$E_\theta T = \frac{n\theta}{n+1} \quad \text{and} \quad E_\theta T^2 = \frac{n\theta^2}{n+2}.$$

So the risk of δ_a is

$$\begin{aligned} R(\theta, \delta_a) &= E_\theta(aT - \theta)^2 \\ &= a^2 E_\theta T^2 - 2a\theta E_\theta T + \theta^2 \\ &= \theta^2 \left(\frac{n}{n+2} a^2 - \frac{2n}{n+1} a + 1 \right). \end{aligned}$$

This is a quadratic function of a minimized when $a = (n+2)/(n+1)$. With this choice for a , $R(\theta, \delta_a) = \theta^2/(n+1)^2$, slightly smaller than the risk $\theta^2/(n^2+2n)$ for the UMVU estimator. With squared error loss, the risk of an arbitrary estimator δ can be written as

$$R(\theta, \delta) = E_\theta (\delta - g(\theta))^2 = \text{Var}_\theta(\delta) + b^2(\theta, \delta),$$

where $b(\theta, \delta) = E_\theta \delta - g(\theta)$ is the bias of δ . In this example, the biased estimator δ_a has smaller variance than the UMVU estimator, which more than compensates for a small amount of additional risk due to the bias. Possibilities for this kind of trade-off between bias and variance arise fairly often in statistics. In nonparametric curve estimation these trade-offs often play a key role. (See Section 18.1.)

Example 4.7. Suppose X has mass function

$$P_\theta(X = x) = \frac{\theta^x e^{-\theta}}{x!(1 - e^{-\theta})}, \quad x = 1, 2, \dots$$

This is the density for a Poisson distribution truncated to $\{1, 2, \dots\}$. The mass functions for X form an exponential family and X is complete sufficient. Consider estimating $g(\theta) = e^{-\theta}$ (the proportion lost through truncation). If $\delta(X)$ is unbiased, then

$$e^{-\theta} = \sum_{k=1}^{\infty} \frac{\delta(k) \theta^k e^{-\theta}}{k!(1 - e^{-\theta})}, \quad \theta > 0,$$

and so

$$\sum_{k=1}^{\infty} \frac{\delta(k)}{k!} \theta^k = 1 - e^{-\theta} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!} \theta^k, \quad \theta > 0.$$

These power series will agree if and only if they have equal coefficients for θ^k . Hence $\delta(k)$ must be $(-1)^{k+1}$, and the UMVU estimator is $(-1)^{X+1}$, which is 1 when X is odd and -1 when X is even! In this example the only unbiased estimator is absurd.

4.3 Normal One-Sample Problem—Distribution Theory

In this section distributional results related to sampling from a normal distribution are derived. To begin, here are a few useful properties about normal variables. Let $X \sim N(\mu, \sigma^2)$ and take $Z = (X - \mu)/\sigma$.

1. The distribution of Z is standard normal, $Z \sim N(0, 1)$. More generally, if a and b are constants, $aX + b \sim N(a\mu + b, a^2\sigma^2)$.

Proof. The cumulative distribution function of Z is

$$\begin{aligned} P(Z \leq z) &= P\left(\frac{X - \mu}{\sigma} \leq z\right) = P(X \leq \mu + z\sigma) \\ &= \int_{-\infty}^{\mu + z\sigma} \frac{\exp[-(x - \mu)^2/(2\sigma^2)]}{\sqrt{2\pi}\sigma^2} dx = \int_{-\infty}^z \frac{e^{-u^2/2}}{\sqrt{2\pi}} du. \end{aligned}$$

Taking a derivative with respect to z , Z has density $e^{-z^2/2}/\sqrt{2\pi}$ and so $Z \sim N(0, 1)$. The second assertion can be established in a similar fashion. $\square$

2. The moment generating function of Z is $M_Z(u) = e^{u^2/2}$, $u \in \mathbb{R}$.

Proof. Completing the square,

$$M_Z(u) = Ee^{uZ} = \int e^{uz} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz = e^{u^2/2} \int \frac{e^{-(z-u)^2/2}}{\sqrt{2\pi}} dz.$$

The integrand here is the density for $N(u, 1)$, which integrates to one, and the result follows. $\square$

3. The moment generating function of X is

$$M_X(u) = e^{u\mu + u^2\sigma^2/2}, \quad u \in \mathbb{R}.$$

Proof.

$$M_X(u) = Ee^{uX} = Ee^{u(\mu + \sigma Z)} = e^{u\mu} Ee^{u\sigma Z} = e^{u\mu} M_Z(u\sigma) = e^{u\mu + u^2\sigma^2/2}.$$

$\square$

4. If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof.

$$\begin{aligned} M_{X_1+X_2}(u) &= M_{X_1}(u)M_{X_2}(u) = e^{u\mu_1+u^2\sigma_1^2/2}e^{u\mu_2+u^2\sigma_2^2/2} \\ &= e^{u(\mu_1+\mu_2)+u^2(\sigma_1^2+\sigma_2^2)/2}, \quad u \in \mathbb{R}, \end{aligned}$$

which is the moment generating function for $N(\mu, \sigma^2)$ with $\mu = \mu_1 + \mu_2$ and $\sigma^2 = \sigma_1^2 + \sigma_2^2$. So the assertion follows by Lemma 2.7. $\square$

Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$. By Example 2.3, the joint densities parameterized by $\theta = (\mu, \sigma^2)$ form a two-parameter full rank exponential family with complete sufficient statistic $T = (\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$. It is often more convenient to work with statistics

$$\bar{X} = \frac{X_1 + \dots + X_n}{n} \quad \text{and} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

called the sample mean and variance. Using the identity $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$, we have

$$\bar{X} = \frac{T_1}{n} \quad \text{and} \quad S^2 = \frac{T_2 - T_1^2/n}{n-1}, \quad (4.3)$$

or

$$T_1 = n\bar{X} \quad \text{and} \quad T_2 = (n-1)S^2 + n\bar{X}^2. \quad (4.4)$$

This establishes a one-to-one relationship between T and $(\bar{X}, S^2)$. One-to-one relationships preserve sufficiency and completeness, and so $(\bar{X}, S^2)$ is also a complete sufficient statistic.

Iterating Property 4, $X_1 + \dots + X_n \sim N(n\mu, n\sigma^2)$. Dividing by n , by Property 1, $\bar{X} \sim N(\mu, \sigma^2/n)$. We know from Example 3.22 that $\bar{X}$ and S^2 are independent, but the derivation to find the marginal distribution of S^2 is a bit more involved.

The gamma distribution, introduced in Problem 1.26, with parameters $\alpha > 0$ and $\beta > 0$, denoted $\Gamma(\alpha, \beta)$, has density

$$f_{\alpha, \beta}(x) = \begin{cases} \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}, & x > 0; \\ 0, & \text{otherwise,} \end{cases} \quad (4.5)$$

where $\Gamma(\cdot)$ is the gamma function defined as

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx, \quad \Re(\alpha) > 0.$$

Useful properties of the gamma function include

$$\Gamma(\alpha + 1) = \alpha\Gamma(\alpha), \quad \Re(\alpha) > 0$$

(which follows after integration by parts from the definition), $\Gamma(n + 1) = n!$, $n = 1, 2, \dots$, and $\Gamma(1/2) = \sqrt{\pi}$. It is not hard to show that if $X \sim \Gamma(\alpha, 1)$, then $\beta X \sim \Gamma(\alpha, \beta)$. For this reason, β is called a *scale parameter*, and α is called the *shape parameter* for the distribution.

If $X \sim \Gamma(\alpha, \beta)$, then, for $u < 1/\beta$,

$$\begin{aligned} M_X(u) &= Ee^{uX} = \int_0^\infty e^{ux} \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)} dx \\ &= \frac{1}{(1-u\beta)^\alpha} \frac{1}{\Gamma(\alpha)} \int_0^\infty y^{\alpha-1} e^{-y} dy = \frac{1}{(1-u\beta)^\alpha}, \end{aligned}$$

where the change of variables $y = (1-u\beta)x/\beta$ gives the third equality. From this, if $X \sim \Gamma(\alpha_x, \beta)$ and $Y \sim \Gamma(\alpha_y, \beta)$ are independent, then

$$M_{X+Y}(u) = M_X(u)M_Y(u) = \frac{1}{(1-u\beta)^{\alpha_x+\alpha_y}}.$$

This is the moment generating function for $\Gamma(\alpha_x + \alpha_y, \beta)$, and so

$$X + Y \sim \Gamma(\alpha_x + \alpha_y, \beta). \quad (4.6)$$

The chi-square distributions are special cases of the gamma distribution, generally defined as sums of independent squared standard normal variables. If $Z \sim N(0, 1)$, then

$$M_{Z^2}(u) = \int e^{uz^2} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \frac{1}{\sqrt{1-2u}} \int \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \frac{1}{\sqrt{1-2u}},$$

where the change of variables $x = z\sqrt{1-2u}$ gives the second equality. The distribution of Z^2 is called the chi-square distribution on one degree of freedom, denoted χ_1^2 . But the moment generating function for Z^2 just computed is the moment generating function for $\Gamma(1/2, 2)$. So $\chi_1^2 = \Gamma(1/2, 2)$.

Definition 4.8. The chi-square distribution on p degrees of freedom, χ_p^2 , is the distribution of the sum $Z_1^2 + \dots + Z_p^2$ when $Z_1, \dots, Z_p$ are i.i.d. from $N(0, 1)$.

Repeated use of (4.6) shows that

$$\chi_p^2 = \Gamma(p/2, 2),$$

which has moment generating function

$$\frac{1}{(1-2u)^{p/2}}, \quad u < 1/2. \quad (4.7)$$

Returning to sampling from a normal distribution, let $X_1, \dots, X_n$ be i.i.d. from $N(\mu, \sigma^2)$, and define $Z_i = (X_i - \mu)/\sigma$, so that $Z_1, \dots, Z_n$ are i.i.d. from $N(0, 1)$. Then

$$\bar{Z} = \frac{1}{n} \sum_{i=1}^n \frac{X_i - \mu}{\sigma} = \frac{\sum_{i=1}^n X_i - n\mu}{n\sigma} = \frac{\bar{X} - \mu}{\sigma}.$$

Note that $\sqrt{n}\bar{Z} \sim N(0, 1)$, and so $n\bar{Z}^2 \sim \chi_1^2$. Next,

$$\begin{aligned} V &\stackrel{\text{def}}{=} \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \\ &= \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} - \frac{\bar{X} - \mu}{\sigma} \right)^2 \\ &= \sum_{i=1}^n (Z_i - \bar{Z})^2. \end{aligned}$$

Expanding the square,

$$V = \sum_{i=1}^n (Z_i - \bar{Z})^2 = \sum_{i=1}^n (Z_i^2 - 2Z_i\bar{Z} + \bar{Z}) = \sum_{i=1}^n Z_i^2 - n\bar{Z}^2,$$

and thus

$$V + n\bar{Z}^2 = \sum_{i=1}^n Z_i^2 \sim \chi_n^2. \quad (4.8)$$

By Basu's theorem (see Example 3.22), $\bar{X}$ and S^2 are independent. Because $n\bar{Z}^2$ is a function of $\bar{X}$, and V is a function of S^2 , V and $n\bar{Z}^2$ are independent. Using this independence and formula (4.7) for the moment generating function for χ_n^2 , (4.8) implies

$$M_V(u)M_{n\bar{Z}^2}(u) = \frac{1}{(1-2u)^{n/2}}.$$

But $n\bar{Z}^2 \sim \chi_1^2$ with moment generating function $1/\sqrt{1-2u}$, and thus

$$M_V(u) = \frac{1}{(1-2u)^{(n-1)/2}}. \quad (4.9)$$

This is the moment generating function for χ_{n-1}^2 , and thus

$$V = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

This along with $\bar{X} \sim N(\mu, \sigma^2/n)$ implicitly determines the joint distribution of $\bar{X}$ and S^2 , because these two variables are independent.

4.4 Normal One-Sample Problem—Estimation

Results from the last section lead directly to a variety of UMVU estimates. First note that for $n + r > 1$,

$$\begin{aligned} ES^r &= E \left[\frac{\sigma^r}{(n-1)^{r/2}} V^{r/2} \right] \\ &= \frac{\sigma^r}{(n-1)^{r/2}} \int_0^\infty \frac{x^{(r+n-3)/2} e^{-x/2}}{2^{(n-1)/2} \Gamma[(n-1)/2]} dx \\ &= \frac{\sigma^r 2^{r/2} \Gamma[(r+n-1)/2]}{(n-1)^{r/2} \Gamma[(n-1)/2]}. \end{aligned} \quad (4.10)$$

From this,

$$\frac{(n-1)^{r/2} \Gamma[(n-1)/2]}{2^{r/2} \Gamma[(r+n-1)/2]} S^r$$

is an unbiased estimate of σ^r . This estimate is UMVU because it is a function of the complete sufficient statistic $(\bar{X}, S^2)$. In particular, when $r = 2$, S^2 is UMVU for σ^2 . Note that the UMVU estimate for σ is not S , although S is a common and natural choice in practice. By Stirling's formula

$$\frac{(n-1)^{r/2} \Gamma[(n-1)/2]}{2^{r/2} \Gamma[(r+n-1)/2]} = 1 - \frac{r(r-2)}{4n} + O(1/n^2),$$

as $n \rightarrow \infty$.¹ For large n , the bias of S^r as an estimate of σ^r will be slight.

Because $E\bar{X} = \mu$, $\bar{X}$ is the UMVU estimator of μ . However, $\bar{X}^2$ is a biased estimator of μ^2 as

$$E\bar{X}^2 = (E\bar{X})^2 + \text{Var}(\bar{X}) = \mu^2 + \sigma^2/n.$$

The bias can be removed by subtracting an unbiased estimate of σ^2/n . Doing this, $\bar{X}^2 - S^2/n$ is UMVU for μ^2 .

The parameter μ/σ might be interpreted as a signal-to-noise ratio. The unbiased estimate of σ^{-1} given above only depends on S^2 and is independent of $\bar{X}$, the unbiased estimate of μ . Multiplying these estimates together,

$$\frac{\bar{X}\sqrt{2} \Gamma[(n-1)/2]}{S\sqrt{n-1} \Gamma[(n-2)/2]}$$

is UMVU for μ/σ .

The p th quantile for $N(\mu, \sigma^2)$ is a value x such that $P(X_i \leq x) = p$. If Φ is the cumulative distribution function for $N(0, 1)$, then as $Z_i = (X_i - \mu)/\sigma \sim N(0, 1)$,

¹ Here $O(1/n^2)$ represents a remainder bounded in magnitude by some multiple of $1/n^2$. See Section 8.6.

$$P(X_i \leq x) = P\left(Z_i \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

This equals p if $(x - \mu)/\sigma = \Phi^\leftarrow(p)$, and so the p th quantile of $N(\mu, \sigma^2)$ is

$$x = \mu + \sigma\Phi^\leftarrow(p).$$

The UMVU estimate of this quantile is

$$\bar{X} + \frac{\sqrt{n-1}\Gamma[(n-1)/2]}{\sqrt{2}\Gamma(n/2)}S\Phi^\leftarrow(p).$$

4.5 Variance Bounds and Information

From (1.10) and (1.11), the covariance between two random variables X and Y is

$$\text{Cov}(X, Y) = E(X - EX)(Y - EY) = EXY - (EX)(EY).$$

In particular, if either mean, EX or EY , is zero, $\text{Cov}(X, Y) = EXY$. Letting $\sigma_X = \sqrt{\text{Var}(X)}$ and $\sigma_Y = \sqrt{\text{Var}(Y)}$, then because

$$E[(X - EX)\sigma_Y \pm (Y - EY)\sigma_X]^2 = 2\sigma_X\sigma_Y(\sigma_X\sigma_Y \pm \text{Cov}(X, Y)) \geq 0,$$

we have the bound

$$|\text{Cov}(X, Y)| \leq \sigma_X\sigma_Y \text{ or } \text{Cov}^2(X, Y) \leq \text{Var}(X)\text{Var}(Y), \quad (4.11)$$

called the *covariance inequality*.

Using the covariance inequality, if δ is an unbiased estimator of $g(\theta)$ and Ψ is an arbitrary random variable, then

$$\text{Var}_\theta(\delta) \geq \frac{\text{Cov}_\theta^2(\delta, \psi)}{\text{Var}_\theta(\psi)}. \quad (4.12)$$

The right hand side of this inequality involves δ , so this seems rather useless as a bound for the variance of δ . To make headway we need to choose ψ cleverly, so that $\text{Cov}_\theta(\delta, \psi)$ is the same for all δ that are unbiased for $g(\theta)$.

Let $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ be a dominated family with densities p_θ , $\theta \in \Omega \subset \mathbb{R}$. As a starting point, $E_{\theta+\Delta}\delta - E_\theta\delta$ gives the same value $g(\theta + \Delta) - g(\theta)$ for any unbiased δ . Here Δ must be chosen so that $\theta + \Delta \in \Omega$. Next, we write $E_{\theta+\Delta}\delta - E_\theta\delta$ as a covariance under P_θ . To do this we first express $E_{\theta+\Delta}\delta$ as an expectation under P_θ , which is accomplished by introducing a likelihood ratio. This step of the argument involves a key assumption that $p_{\theta+\Delta}(x) = 0$ whenever $p_\theta(x) = 0$. Define $L(x) = p_{\theta+\Delta}(x)/p_\theta(x)$ when $p_\theta(x) > 0$, and $L(x) = 1$, otherwise. (This function L is called a likelihood ratio.) From the assumption,

$$L(x)p_\theta(x) = \frac{p_{\theta+\Delta}(x)}{p_\theta(x)}p_\theta(x) = p_{\theta+\Delta}(x), \quad \text{a.e. } x,$$

and so, for any function h integrable under $P_{\theta+\Delta}$,

$$E_{\theta+\Delta}h(X) = \int h p_{\theta+\Delta} d\mu = \int h L p_\theta d\mu = E_\theta L(X)h(X).$$

Taking $h = 1$, $E_\theta L = 1$; and taking $h = \delta$, $E_{\theta+\Delta}\delta = E_\theta L\delta$. So if we define

$$\psi(X) = L(X) - 1,$$

then $E_\theta\psi = 0$ and

$$E_{\theta+\Delta}\delta - E_\theta\delta = E_\theta L\delta - E_\theta\delta = E_\theta\psi\delta = \text{Cov}_\theta(\delta, \psi).$$

Thus

$$\text{Cov}_\theta(\delta, \psi) = g(\theta + \Delta) - g(\theta)$$

for any unbiased estimator δ . With this choice for ψ , (4.12) gives

$$\text{Var}_\theta(\delta) \geq \frac{[g(\theta + \Delta) - g(\theta)]^2}{\text{Var}_\theta(\psi)} = \frac{[g(\theta + \Delta) - g(\theta)]^2}{E_\theta \left(\frac{p_{\theta+\Delta}(X)}{p_\theta(X)} - 1 \right)^2}, \quad (4.13)$$

called the Hammersley–Chapman–Robbins inequality.

Under suitable regularity, the dominated convergence theorem can be used to show that the lower bound in (4.13), which can be written as

$$\frac{\left[\frac{g(\theta + \Delta) - g(\theta)}{\Delta} \right]^2}{E_\theta \left(\frac{[p_{\theta+\Delta}(X) - p_\theta(X)]/\Delta}{p_\theta(X)} \right)^2},$$

converges to

$$\frac{[g'(\theta)]^2}{E_\theta \left(\frac{\partial p_\theta(X)/\partial \theta}{p_\theta(X)} \right)^2}$$

as $\Delta \rightarrow 0$. The denominator here is called *Fisher information*, denoted $I(\theta)$, and given by

$$I(\theta) = E_\theta \left(\frac{\partial \log p_\theta(X)}{\partial \theta} \right)^2. \quad (4.14)$$

With enough regularity to interchange integration and differentiation,

$$\begin{aligned}
 0 &= \frac{\partial}{\partial \theta} 1 = \frac{\partial}{\partial \theta} \int p_\theta(x) d\mu(x) = \int \frac{\partial}{\partial \theta} p_\theta(x) d\mu(x) \\
 &= \int \frac{\partial \log p_\theta(x)}{\partial \theta} p_\theta(x) d\mu(x) = E_\theta \frac{\partial \log p_\theta(X)}{\partial \theta},
 \end{aligned}$$

and so

$$I(\theta) = \text{Var}_\theta \left(\frac{\partial \log p_\theta(X)}{\partial \theta} \right). \quad (4.15)$$

If we can pass two partial derivatives with respect to θ inside the integral $\int p_\theta d\mu = 1$, then

$$\int \frac{\partial^2 p_\theta(x)}{\partial \theta^2} d\mu(x) = E_\theta \left[\frac{\partial^2 p_\theta(X)/\partial \theta^2}{p_\theta(X)} \right] = 0.$$

From this, inasmuch as

$$\begin{aligned}
 \frac{\partial^2 \log p_\theta(X)}{\partial \theta^2} &= \frac{\partial^2 p_\theta(X)/\partial \theta^2}{p_\theta(X)} - \left(\frac{\partial \log p_\theta(X)}{\partial \theta} \right)^2, \\
 I(\theta) &= -E_\theta \frac{\partial^2 \log p_\theta(X)}{\partial \theta^2}.
 \end{aligned} \quad (4.16)$$

For calculations, this formula is often more convenient than (4.14).

A lower bound based on Fisher information can be derived in much the same way as the Hammersley–Chapman–Robbins inequality, but tends to involve differentiation under an integral sign. Let δ have mean $g(\theta) = E_\theta \delta$ and take $\psi = \partial \log p_\theta / \partial \theta$. With sufficient regularity,

$$g'(\theta) = \frac{\partial}{\partial \theta} \int \delta p_\theta d\mu = \int \delta \frac{\partial}{\partial \theta} p_\theta d\mu = \int \delta \psi p_\theta d\mu,$$

or

$$g'(\theta) = E_\theta \delta \psi. \quad (4.17)$$

In a given application, this might be established using dominated convergence. If δ is identically one, then $g(\theta) = 1$, $g'(\theta) = 0$, and we anticipate $E_\theta \psi = 0$. Then (4.17) shows that $\text{Cov}_\theta(\delta, \psi) = g'(\theta)$. Using this in (4.12) we have the following result.

Theorem 4.9. *Let $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ be a dominated family with Ω an open set in $\mathbb{R}$ and densities p_θ differentiable with respect to θ . If $E_\theta \psi = 0$, $E_\theta \delta^2 < \infty$, and (4.17) hold for all $\theta \in \Omega$, then*

$$\text{Var}_\theta(\delta) \geq \frac{[g'(\theta)]^2}{I(\theta)}, \quad \theta \in \Omega.$$

This result is called the Cramér–Rao, or information, bound. The regularity condition (4.17) is troublesome. It involves the estimator δ , thus the

theorem leaves open the possibility that some estimators, not satisfying (4.17), may have variance below the stated bound. This has been addressed in various ways. Under very weak conditions Woodroffe and Simons (1983) show that the bound holds for any estimator δ for almost all θ . Other authors impose more restrictive conditions on the model $\mathcal{P}$, but show that the bound holds for all δ at all $\theta \in \Omega$.

Suppose $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ is a dominated family with densities p_θ and Fisher information I . If h is a one-to-one function from Ξ to Ω , then the family $\mathcal{P}$ can be reparameterized as $\tilde{\mathcal{P}} = \{Q_\xi : \xi \in \Xi\}$ with the identification $Q_\xi = P_{h(\xi)}$. Then Q_ξ has density $q_\xi = p_{h(\xi)}$. Letting $\theta = h(\xi)$, by the chain rule, Fisher information $\tilde{I}$ for the reparameterized family $\tilde{\mathcal{P}}$ is given by

$$\begin{aligned}\tilde{I}(\xi) &= \tilde{E}_\xi \left(\frac{\partial \log q_\xi(X)}{\partial \xi} \right)^2 = \tilde{E}_\xi \left(\frac{\partial \log p_{h(\xi)}(X)}{\partial \xi} \right)^2 \\ &= [h'(\xi)]^2 E_\theta \left(\frac{\partial \log p_\theta(X)}{\partial \theta} \right)^2 = [h'(\xi)]^2 I(\theta).\end{aligned}\quad (4.18)$$

Example 4.10. Exponential Families. Let $\mathcal{P}$ be a one-parameter exponential family in canonical form with densities p_η given by

$$p_\eta(x) = \exp[\eta T(x) - A(\eta)] h(x).$$

Then

$$\frac{\partial \log p_\eta(X)}{\partial \eta} = T - A'(\eta),$$

and so by (4.15),

$$I(\eta) = \text{Var}_\eta(T - A'(\eta)) = \text{Var}_\eta(T) = A''(\eta).$$

Because

$$\frac{\partial^2 \log p_\eta(X)}{\partial \eta^2} = -A''(\eta),$$

this formula for $I(\eta)$ also follows immediately from (4.16). If the family is parameterized instead by $\mu = A'(\eta) = E_\eta T$, then by (4.18)

$$A''(\eta) = I(\mu) [A''(\eta)]^2,$$

and so, because $A''(\eta) = \text{Var}(T)$,

$$I(\mu) = \frac{1}{\text{Var}_\mu T}.$$

Note that because T is UMVU for μ , the lower bound $\text{Var}_\mu(\delta) \geq 1/I(\mu)$ for an unbiased estimator δ of μ is sharp in this example.

Example 4.11. Location Families. Suppose ϵ is an absolutely continuous random variable with density f . The family of distributions $\mathcal{P} = \{P_\theta : \theta \in \mathbb{R}\}$ with P_θ the distribution of $\theta + \epsilon$ is called a *location family*. Using a change of variables $x = \theta + e$,

$$\begin{aligned} \int g(x) dP_\theta(x) &= E_\theta g(X) = E g(\theta + \epsilon) \\ &= \int g(\theta + e) f(e) de = \int g(x) f(x - \theta) dx, \end{aligned}$$

and so P_θ has density $p_\theta(x) = f(x - \theta)$. Fisher information for this family is given by

$$\begin{aligned} I(\theta) &= E_\theta \left(\frac{\partial \log f(X - \theta)}{\partial \theta} \right)^2 = E_\theta \left(-\frac{f'(X - \theta)}{f(X - \theta)} \right)^2 \\ &= E \left(\frac{f'(\epsilon)}{f(\epsilon)} \right)^2 = \int \frac{[f'(x)]^2}{f(x)} dx. \end{aligned}$$

So for location families, $I(\theta)$ is constant and does not vary with θ .

If two (or more) independent vectors are observed, then the total Fisher information is the sum of the Fisher information provided by the individual observations. To see this, suppose X and Y are independent, and that X has density p_θ and Y has density q_θ (dominating measures for the distributions of X and Y can be different). Then by (4.15), the Fisher information observing X is

$$I_X(\theta) = \text{Var}_\theta \left(\frac{\partial \log p_\theta(X)}{\partial \theta} \right),$$

and the Fisher information observing Y is

$$I_Y(\theta) = \text{Var}_\theta \left(\frac{\partial \log q_\theta(Y)}{\partial \theta} \right).$$

As X and Y are independent, their joint density is $p_\theta(x)q_\theta(y)$, and Fisher information observing both vectors X and Y is

$$\begin{aligned} I_{X,Y}(\theta) &= \text{Var}_\theta \left(\frac{\partial \log [p_\theta(X)q_\theta(Y)]}{\partial \theta} \right) \\ &= \text{Var}_\theta \left(\frac{\partial \log p_\theta(X)}{\partial \theta} + \frac{\partial \log q_\theta(Y)}{\partial \theta} \right) \\ &= \text{Var}_\theta \left(\frac{\partial \log p_\theta(X)}{\partial \theta} \right) + \text{Var}_\theta \left(\frac{\partial \log q_\theta(Y)}{\partial \theta} \right) \\ &= I_X(\theta) + I_Y(\theta). \end{aligned}$$

Iterating this, the Fisher information for a random sample of n observations will be $nI(\theta)$ if $I(\theta)$ denotes the Fisher information for a single observation.

4.6 Variance Bounds in Higher Dimensions

When the parameter θ takes values in $\mathbb{R}^s$, Fisher information will be a matrix, defined in regular cases by

$$\begin{aligned} I(\theta)_{i,j} &= E_\theta \left[\frac{\partial \log p_\theta(X)}{\partial \theta_i} \frac{\partial \log p_\theta(X)}{\partial \theta_j} \right] \\ &= \text{Cov}_\theta \left(\frac{\partial \log p_\theta(X)}{\partial \theta_i}, \frac{\partial \log p_\theta(X)}{\partial \theta_j} \right) \\ &= -E_\theta \left[\frac{\partial^2 \log p_\theta(X)}{\partial \theta_i \partial \theta_j} \right]. \end{aligned}$$

The first two lines here are equal because

$$E_\theta \nabla_\theta \log p_\theta(X) = 0,$$

and, as before, the third formula requires extra regularity necessary to pass a second derivative inside an integral. Using matrix notation,

$$\begin{aligned} I(\theta) &= E_\theta (\nabla_\theta \log p_\theta(X)) (\nabla_\theta \log p_\theta(X))' \\ &= \text{Cov}_\theta (\nabla_\theta \log p_\theta(X)) = -E_\theta \nabla_\theta^2 \log p_\theta(X), \end{aligned}$$

where ∇_θ is the gradient with respect to θ , ∇_θ^2 is the Hessian matrix of second order derivatives, and prime denotes transpose. The lower bound for the variance of an unbiased estimator δ of $g(\theta)$, where $g: \Omega \rightarrow \mathbb{R}$, is

$$\text{Var}_\theta(\delta) \geq \nabla g(\theta)' I^{-1}(\theta) \nabla g(\theta).$$

Example 4.12. Exponential Families. If $\mathcal{P}$ is an s -parameter exponential family in canonical form with densities

$$p_\eta(x) = \exp[\eta \cdot T(x) - A(\eta)] h(x),$$

then

$$\frac{\partial^2 \log p_\eta(X)}{\partial \eta_i \partial \eta_j} = -\frac{\partial^2 A(\eta)}{\partial \eta_i \partial \eta_j}.$$

Thus

$$I(\eta)_{i,j} = \frac{\partial^2 A(\eta)}{\partial \eta_i \partial \eta_j}.$$

This can be written more succinctly as

$$I(\eta) = \nabla^2 A(\eta).$$

The final formula in this section is a multivariate extension of (4.18). As before, let $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ be a dominated family with densities p_θ and Fisher information I , but now Ω is a subset of $\mathbb{R}^s$. Let h be a differentiable one-to-one function from Ξ to Ω and introduce the family $\tilde{\mathcal{P}} = \{Q_\xi : \xi \in \Xi\}$ with $Q_\xi = P_{h(\xi)}$. The density for Q_ξ is $q_\xi = p_{h(\xi)}$, and by the chain rule,

$$\frac{\partial \log q_\xi(X)}{\partial \xi_i} = \sum_j \frac{\partial \log p_\theta(X)}{\partial \theta_j} \frac{\partial h_j(\xi)}{\partial \xi_i},$$

where $\theta = h(\xi)$. If Dh represents the matrix of partial derivatives of h given by

$$[Dh(\xi)]_{i,j} = \frac{\partial h_j(\xi)}{\partial \xi_i},$$

then $\partial \log q_\xi(X) / \partial \xi_i$ is the i th entry of $[Dh(\xi)]' \nabla_\theta \log p_\theta(X)$. So

$$\nabla_\xi \log q_\xi(X) = [Dh(\xi)]' \nabla_\theta \log p_\theta(X)$$

and

$$\begin{aligned} \tilde{I}(\xi) &= \tilde{E}_\xi [\nabla_\xi \log q_\xi(X)] [\nabla_\xi \log q_\xi(X)]' \\ &= E_\theta [Dh(\xi)]' [\nabla_\theta \log p_\theta(X)] [\nabla_\theta \log p_\theta(X)]' [Dh(\xi)] \\ &= [Dh(\xi)]' I(\theta) [Dh(\xi)]. \end{aligned}$$

4.7 Problems²

- *1. Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent variables with the X_i a random sample from an exponential distribution with failure rate λ_x , and the Y_j a random sample from an exponential distribution with failure rate λ_y .
 - a) Determine the UMVU estimator of λ_x/λ_y .
 - b) Under squared error loss, find the best estimator of λ_x/λ_y of form $\delta = c\bar{Y}/\bar{X}$.
 - c) Find the UMVU estimator of $e^{-\lambda_x} = P(X_1 > 1)$.
- *2. Let $X_1, \dots, X_n$ be a random sample from $N(\mu_x, \sigma^2)$, and let $Y_1, \dots, Y_m$ be an independent random sample from $N(\mu_y, 2\sigma^2)$, with μ_x , μ_y , and σ^2 all unknown parameters.
 - a) Find a complete sufficient statistic.
 - b) Determine the UMVU estimator of σ^2 . Hint: Find a linear combination L of $S_x^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n-1)$ and $S_y^2 = \sum_{j=1}^m (Y_j - \bar{Y})^2 / (m-1)$ so that $(\bar{X}, \bar{Y}, L)$ is complete sufficient.

² Solutions to the starred problems are given at the back of the book.

- c) Find a UMVU estimator of $(\mu_x - \mu_y)^2$.
- d) Suppose we know the $\mu_y = 3\mu_x$. What is the UMVU estimator of μ_x ?
- *3. Let $X_1, \dots, X_n$ be a random sample from the Poisson distribution with mean λ . Find the UMVU for $\cos \lambda$. (Hint: For Taylor expansion, the identity $\cos \lambda = (e^{i\lambda} + e^{-i\lambda})/2$ may be useful.)
- *4. Let $X_1, \dots, X_n$ be independent normal variables, each with unit variance, and with $EX_i = \alpha t_i + \beta t_i^2$, $i = 1, \dots, n$, where α and β are unknown parameters and $t_1, \dots, t_n$ are known constants. Find UMVU estimators of α and β .
- *5. Let $X_1, \dots, X_n$ be i.i.d. from some distribution Q_θ , and let $\bar{X} = (X_1 + \dots + X_n)/n$ be the sample average.
- a) Show that $S^2 = \sum (X_i - \bar{X})^2 / (n-1)$ is unbiased for $\sigma^2 = \sigma^2(\theta) = \text{Var}_\theta(X_i)$.
- b) If Q_θ is the Bernoulli distribution with success probability θ , show that S^2 from (a) is UMVU.
- c) If Q_θ is the exponential distribution with failure rate θ , find the UMVU estimator of $\sigma^2 = 1/\theta^2$. Give a formula for $E_\theta[X_i^2 | \bar{X} = c]$ in this case.
- *6. Suppose δ is a UMVU estimator of $g(\theta)$; U is an unbiased estimator of zero, $E_\theta U = 0$, $\theta \in \Omega$; and that δ and U both have finite variances for all $\theta \in \Omega$. Show that U and δ are uncorrelated, $E_\theta U \delta = 0$, $\theta \in \Omega$.
- *7. Suppose δ_1 is a UMVU estimator of $g_1(\theta)$, δ_2 is UMVU estimator of $g_2(\theta)$, and that δ_1 and δ_2 both have finite variance for all θ . Show that $\delta_1 + \delta_2$ is UMVU for $g_1(\theta) + g_2(\theta)$. Hint: Use the result in the previous problem.
- *8. Let $X_1, \dots, X_n$ be i.i.d. absolutely continuous variables with common density f_θ , $\theta > 0$, given by

$$f_\theta(x) = \begin{cases} \theta/x^2, & x > \theta; \\ 0, & x \leq \theta. \end{cases}$$

Find the UMVU estimator for $g(\theta)$ if $g(\theta)/\theta^n \rightarrow 0$ as $\theta \rightarrow \infty$ and g is differentiable.

9. Let X be a single observation from a Poisson distribution with mean λ . Determine the UMVU estimator for

$$e^{-2\lambda} = [P_\lambda(X = 0)]^2.$$

- *10. Suppose X is an exponential variable with density $p_\theta(x) = \theta e^{-\theta x}$, $x > 0$; $p_\theta(x) = 0$, otherwise. Find the UMVU estimator for $1/(1 + \theta)$.
- *11. Let $X_1, \dots, X_3$ be i.i.d. geometric variables with common mass function $f_\theta(x) = P_\theta(X_i = x) = \theta(1 - \theta)^x$, $x = 0, 1, \dots$. Find the UMVU estimator of θ^2 .
- *12. Let X be a single observation, absolutely continuous with density

$$p_{\theta}(x) = \begin{cases} \frac{1}{2}(1 + \theta x), & |x| < 1; \\ 0, & |x| \geq 1. \end{cases}$$

Here $\theta \in [-1, 1]$ is an unknown parameter.

- a) Find a constant a so that aX is unbiased for θ .
 - b) Show that $b = E_{\theta}|X|$ is independent of θ .
 - c) Let θ_0 be a fixed parameter value in $[-1, 1]$. Determine the constant $c = c_{\theta_0}$ that minimizes the variance of the unbiased estimator $aX + c(|X| - b)$ when $\theta = \theta_0$. Is aX uniformly minimum variance unbiased?
13. Let $X_1, \dots, X_m$ be i.i.d. from a Poisson distribution with parameter λ_x and let $Y_1, \dots, Y_n$ be i.i.d. from a Poisson distribution with parameter λ_y , with all $n + m$ variables independent.
- a) Find the UMVU of $(\lambda_x - \lambda_y)^2$.
 - b) Give a formula for the chance X_i is odd, and find the UMVU estimator of this parameter.
14. Let $X_1, \dots, X_n$ be i.i.d. from an arbitrary discrete distribution on $\{0, 1, 2\}$. Let $T_1 = X_1 + \dots + X_n$ and $T_2 = X_1^2 + \dots + X_n^2$.
- a) Show that $T = (T_1, T_2)$ is complete sufficient.
 - b) Let $\mu = EX_i$. Find the UMVU of μ^3 .
15. Let $X_1, \dots, X_n$ be i.i.d. absolutely continuous random variables with common marginal density f_{θ} given by

$$f_{\theta}(x) = \begin{cases} e^{\theta-x}, & x \geq \theta; \\ 0, & x < \theta. \end{cases}$$

Find UMVU estimators for θ and θ^2 .

16. Let X_1 and X_2 be independent discrete random variables with common mass function

$$P(X_i = x) = -\frac{\theta^x}{x \log(1 - \theta)}, \quad x = 1, 2, \dots,$$

where $\theta \in (0, 1)$.

- a) Find the mean and variance of X_1 .
 - b) Find the UMVU of $\theta / \log(1 - \theta)$.
17. Let $X_1, \dots, X_n$ be i.i.d. absolutely continuous variables with common density f_{θ} , $\theta \in \mathbb{R}$, given by

$$f_{\theta}(x) = \begin{cases} \frac{\phi(x)}{\Phi(\theta)}, & x < \theta; \\ 0, & x \geq \theta. \end{cases}$$

(This is the density for the standard normal distribution truncated above at θ .)

- a) Derive a formula for the UMVU for $g(\theta)$. (Assume g is differentiable and behaves reasonably as $\theta \rightarrow \pm\infty$.)

- b) If $n = 3$ and the observed data are -2.3 , -1.2 , and 0 , what is the estimate for θ^2 ?
18. Let X_1 and X_2 be i.i.d. discrete variables with common mass function

$$f_\theta(x) = P_\theta(X_i = x) = (x+1)\theta^2(1-\theta)^x, \quad x = 0, 1, \dots,$$

where $\theta \in (0, 1)$.

- a) Compute $E[1/(X_i + 1)]$.
 - b) Find the mass function for $X_1 + X_2$.
 - c) Use conditioning to find the UMVU for θ .
19. Let X have a binomial distribution with n trials and success probability $\theta \in (0, 1)$. If $m \leq n$, find the UMVU estimator of θ^m .
20. Let $X_1, \dots, X_n$ be i.i.d. and absolutely continuous with common marginal density f_θ given by

$$f_\theta(x) = \begin{cases} 2x/\theta^2, & 0 < x < \theta; \\ 0, & \text{otherwise,} \end{cases}$$

where $\theta > 0$ is an unknown parameter. Find the UMVU estimator of $g(\theta)$ if g is differentiable and $\theta^{2n}g(\theta) \rightarrow 0$ as $\theta \downarrow 0$.

21. Let $X_1, \dots, X_n$ be i.i.d. from an exponential distribution with density

$$f_\theta(x) = \begin{cases} \theta e^{-\theta x}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Find UMVU estimators for θ and θ^2 .

22. Suppose $X_1, \dots, X_n$ are independent with $X_j \sim N(0, j\theta^2)$, $j = 1, \dots, n$. Find the UMVU estimator of θ .
23. For $\theta > 0$, let

$$\Delta_\theta = \{(x, y) \in \mathbb{R}^2 : x > 0, y > 0, x + y < \theta\},$$

the interior of a triangle. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. from the uniform distribution on Δ_θ , so their common density is $21_{\Delta_\theta}/\theta^2$.

- a) Find a complete sufficient statistic T .
 - b) Find the UMVU estimators of θ and $\cos \theta$.
- *24. In the normal one-sample problem, the statistic $t = \sqrt{n}\bar{X}/S$ has the noncentral t -distribution on $n - 1$ degrees of freedom and noncentrality parameter $\delta = \sqrt{n}\mu/\sigma$. Use our results on distribution theory for the one-sample problem to find the mean and variance of t .
25. Let $X_1, \dots, X_n$ be i.i.d. from $N(\mu, \sigma^2)$, with μ and σ both unknown.
- a) Find the UMVU estimator of μ^3 .
 - b) Find the UMVU estimator of μ^2/σ^2 .
26. Let $X_1, \dots, X_n$ be independent with

$$X_i \sim N(m_i\mu, m_i\sigma^2), \quad i = 1, \dots, n,$$

where $m_1, \dots, m_n$ are known constants and μ and σ are unknown parameters. (These sort of data would arise if i.i.d. variables from $N(\mu, \sigma^2)$ were divided into groups, with the i th group having m_i observations, and the observed data are totals for the n groups.)

- a) Find UMVU estimators for μ and σ^2 .
 - b) Show that the estimators in part (a) are independent.
27. Let Z_1 and Z_2 be independent standard normal random variables. Find $E\sqrt{|Z_1/Z_2|}$.
- *28. Let $X_1, \dots, X_n$ be i.i.d. from the uniform distribution on $(0, \theta)$.
- a) Use the Hammersley–Chapman–Robbins inequality to find a lower bound for the variance of an unbiased estimator of θ . This bound will depend on Δ . Note that Δ cannot vary freely but must lie in a suitable set.
 - b) In principle, the best lower bound can be found taking the supremum over Δ . This calculation cannot be done explicitly, but an approximation is possible. Suppose $\Delta = -c\theta/n$. Show that the lower bound for the variance can be written as $\theta^2 g_n(c)/n^2$. Determine $g(c) = \lim_{n \rightarrow \infty} g_n(c)$.
 - c) Find the value c_0 that maximizes $g(c)$ over $c \in (0, 1)$ and give an approximate lower bound for the variance of δ . (The value c_0 cannot be found explicitly, but you should be able to come up with a numerical value.)
29. Determine the Fisher information $I(\theta)$ for the density $f_\theta(x) = (1 + \theta x)/2$, $x \in (-1, 1)$, $f_\theta(x) = 0$, $x \notin (-1, 1)$.
- *30. Suppose $X_1, \dots, X_n$ are independent with $X_i \sim N(\alpha + \beta t_i, 1)$, $i = 1, \dots, n$, where $t_1, \dots, t_n$ are known constants and α, β are unknown parameters.
- a) Find the Fisher information matrix $I(\alpha, \beta)$.
 - b) Give a lower bound for the variance of an unbiased estimator of α .
 - c) Suppose we know the value of β . Give a lower bound for the variance of an unbiased estimator of α in this case.
 - d) Compare the estimators in parts (b) and (c). When are the bounds the same? If the bounds are different, which is larger?
 - e) Give a lower bound for the variance of an unbiased estimator of the product $\alpha\beta$.
- *31. Find the Fisher information for the Cauchy location family with densities p_θ given by

$$p_\theta(x) = \frac{1}{\pi[(x - \theta)^2 + 1]}.$$

Also, what is the Fisher information for θ^3 ?

- *32. Suppose X has a Poisson distribution with mean θ^2 , so the parameter θ is the square root of the usual parameter $\lambda = EX$. Show that the Fisher information $I(\theta)$ is constant.

- *33. Consider the exponential distribution with failure rate λ . Find a function h defining a new parameter $\theta = h(\lambda)$ so that Fisher information $I(\theta)$ is constant.
- *34. Consider an autoregressive model in which $X_1 \sim N(\theta, \sigma^2/(1 - \rho^2))$ and the conditional distribution of X_{j+1} given $X_1 = x_1, \dots, X_j = x_j$, is $N(\theta + \rho(x_j - \theta), \sigma^2)$, $j = 1, \dots, n - 1$.
- Find the Fisher information matrix, $I(\theta, \sigma)$.
 - Give a lower bound for the variance of an unbiased estimator of θ .
 - Show that the sample average $\bar{X} = (X_1 + \dots + X_n)/n$ is an unbiased estimator of θ , compute its variance, and compare its variance with the lower bound. Hint: Define $\epsilon_j = X_j - \theta$ and $\eta_{j+1} = \epsilon_{j+1} - \epsilon_j$. Use smoothing to argue that $\eta_2, \dots, \eta_n$ are i.i.d. $N(0, \sigma^2)$ and are independent of ϵ_1 . Similarly, X_i is independent of $\eta_{i+1}, \eta_{i+2}, \dots$. Use these facts to find first $\text{Var}(X_2) = \text{Var}(\epsilon_2)$, then $\text{Var}(X_3)$, $\text{Var}(X_4)$, $\dots$. Finally, find $\text{Cov}(X_{i+1}, X_i)$, $\text{nCov}(X_{i+2}, X_i)$, and so on.
35. Consider the binomial distribution with n trials and success probability p . Find a function h defining a new parameter $\theta = h(p)$ so that Fisher information $I(\theta)$ is constant.
36. Let $X_1, \dots, X_n$ be i.i.d. with common density $f_\theta(x) = e^{\theta-x}$, $x > \theta$, $f_\theta(x) = 0$, otherwise.
- Find lower bounds for the variance of an unbiased estimator of θ using the Hammersley–Chapman–Robbins inequality. These bounds will depend on the choice of Δ .
 - What choice of Δ gives the best (largest) lower bound?
37. Suppose X has a Poisson distribution with mean λ , and that given $X = n$, Y is Poisson with mean $n\theta$.
- Find the Fisher information matrix.
 - Derive a formula for $\mu_Y = EY$.
 - Find a lower bound for the variance of an unbiased estimator of μ_Y .
 - Compare the bound in part (c) with the variance of Y .
38. Suppose X has a geometric distribution with parameter θ , so $P(X = x) = \theta(1 - \theta)^x$, $x = 0, 1, \dots$, and that given $X = n$, Y is binomial with x trials and success probability p .
- Find the Fisher information matrix.
 - Give a lower bound for the variance of an unbiased estimator of $\mu_Y = EY$. Compare the lower bound with $\text{Var}(Y)$.
39. Let X have a “triangular” shaped density given by

$$f_\theta(x) = \begin{cases} 2(\theta - x)/\theta^2, & x \in (0, \theta); \\ 0, & \text{otherwise.} \end{cases}$$

- Use the Hammersley–Chapman–Robbins inequality to derive lower bounds for the variance of an unbiased estimator of θ based on a single observation X . These bounds will depend on the choice of Δ .

- b) What is in fact the smallest possible variance for an unbiased estimator $\delta(X)$ of θ ? Compare this value with the lower bounds in part (a).
40. Let X have a geometric distribution with success probability θ , so $P_\theta(X = x) = \theta(1 - \theta)^x$, $x = 0, 1, \dots$. What is the smallest possible variance for an unbiased estimator $\delta(X)$ of θ ? Compare this variance with the Cramér–Rao lower bound in Theorem 4.9.
41. Let $X_1, \dots, X_n$ be i.i.d. random variables (angles) from the von Mises distribution with Lebesgue density

$$p_\theta(x) = \begin{cases} \frac{\exp\{\theta_1 \sin x + \theta_2 \cos x\}}{2\pi I_0(\|\theta\|)}, & x \in (0, 2\pi); \\ 0, & \text{otherwise.} \end{cases}$$

Here $\|\theta\|$ denotes the Euclidean length of θ , $\|\theta\| = (\theta_1^2 + \theta_2^2)^{1/2}$, and the function I_0 is a modified Bessel function.

- a) Find the Fisher information matrix, expressed using I_0 and its derivatives.
- b) Give a lower bound for the variance of an unbiased estimator of $\|\theta\|$.
42. Let $\theta = (\alpha, \lambda)$ and let P_θ denote the gamma distribution with shape parameter α and scale $1/\lambda$. So P_θ has density

$$p_\theta(x) = \begin{cases} \frac{\lambda^\alpha x^{\alpha-1} e^{-x\lambda}}{\Gamma(\alpha)}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

- a) Find the Fisher information matrix $I(\theta)$, expressed using the “psi” function $\psi \stackrel{\text{def}}{=} \Gamma'/\Gamma$ and its derivatives.
- b) What is the Cramér–Rao lower bound for the variance of an unbiased estimator of $\alpha + \lambda$?
- c) Find the mean μ and variance σ^2 for P_θ . Show that there is a one-to-one correspondence between θ and (μ, σ^2) .
- d) Find the Fisher information matrix if the family of gamma distributions is parameterized by (μ, σ^2) , instead of θ .

